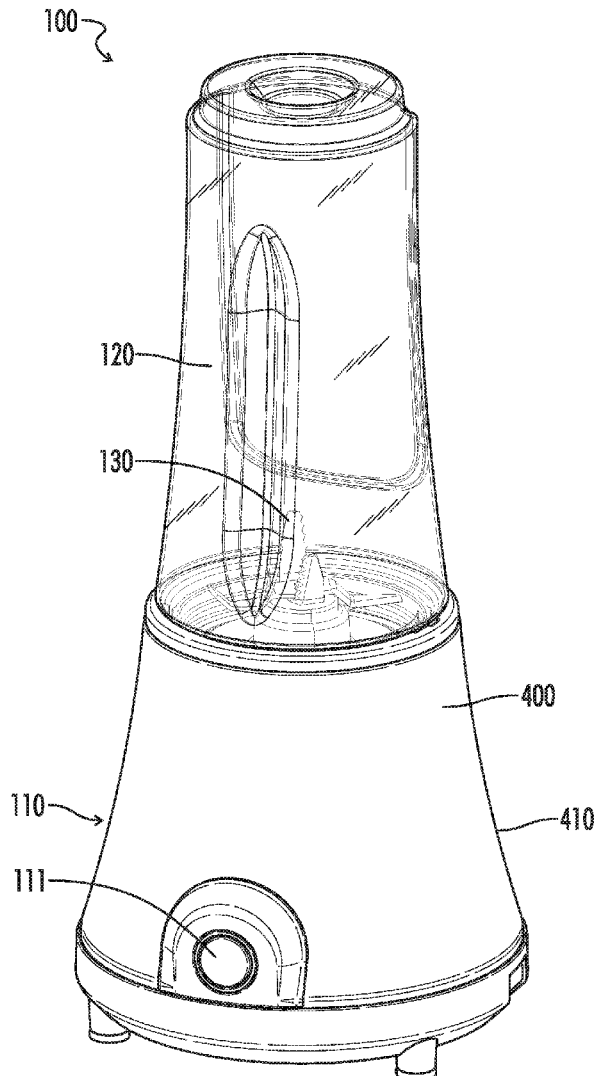


(43) **Pub. Date:** **Nov. 5, 2020**



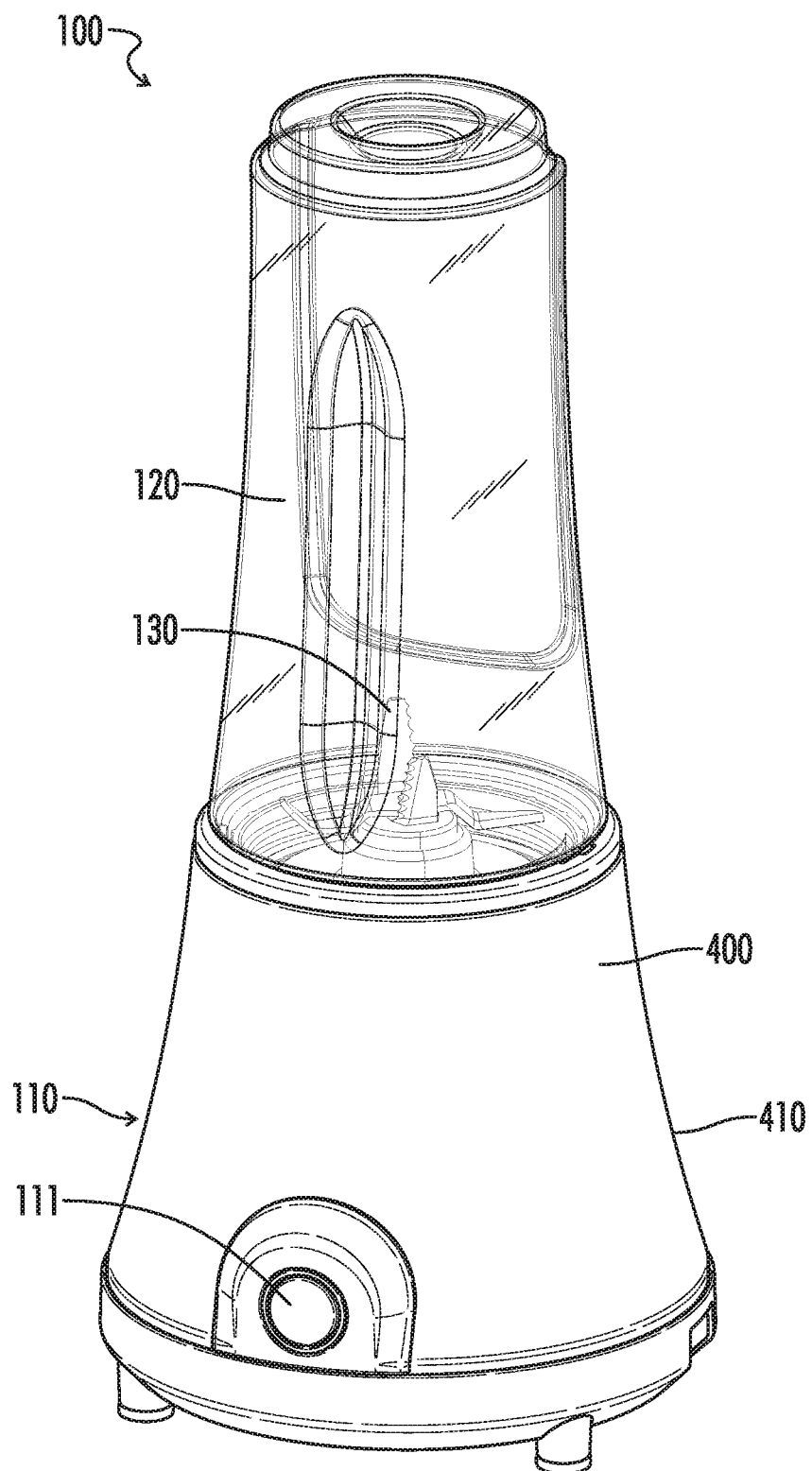


FIG. 1

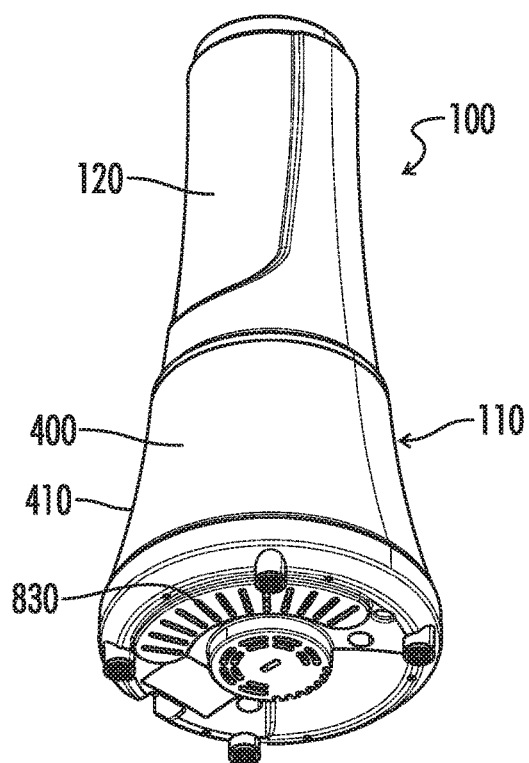


FIG. 2A

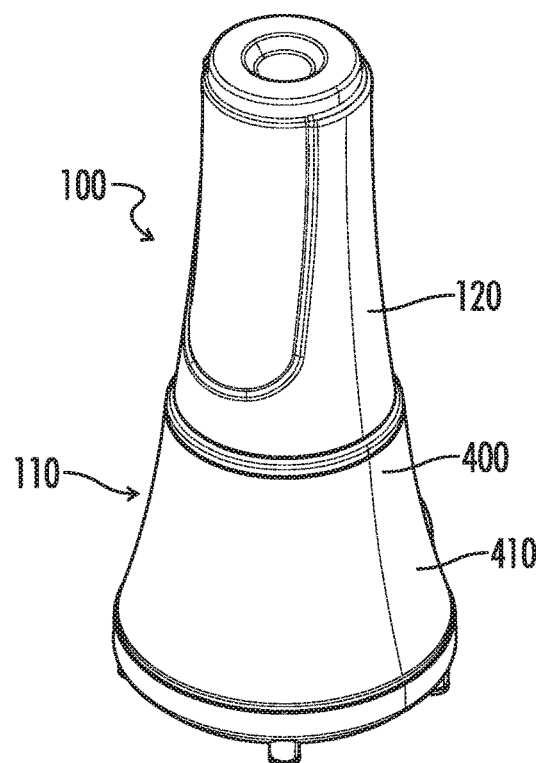


FIG. 2B

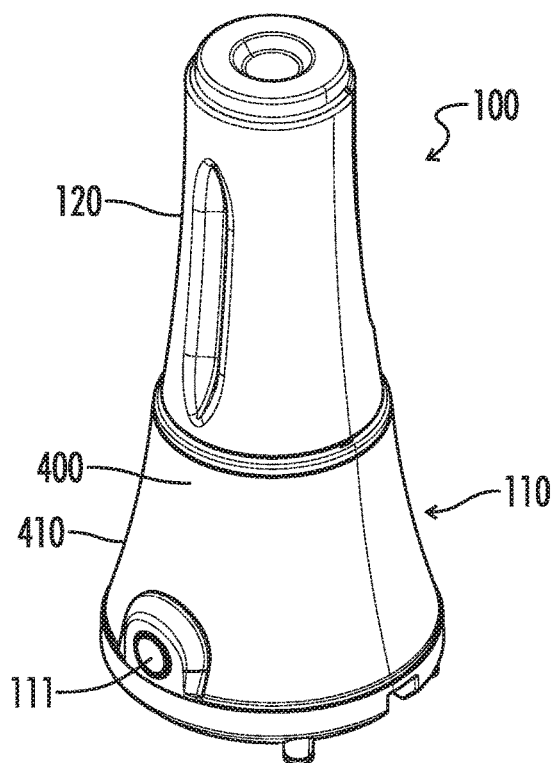


FIG. 2C

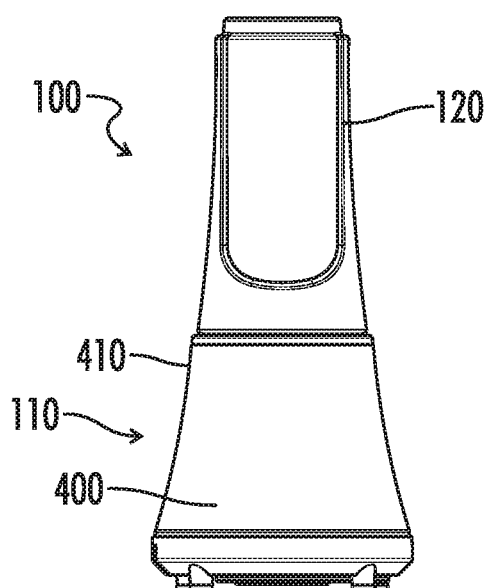


FIG. 3A

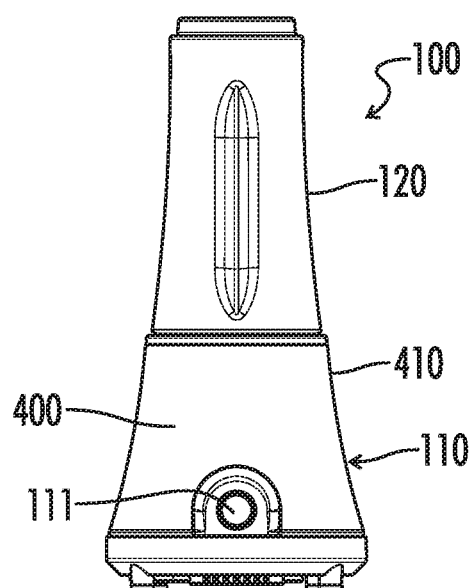


FIG. 3B

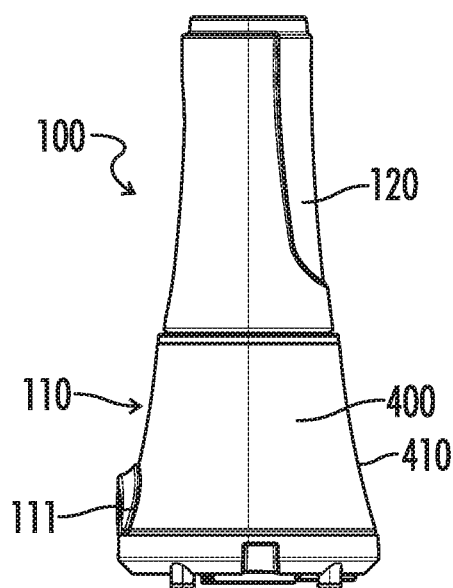


FIG. 3C

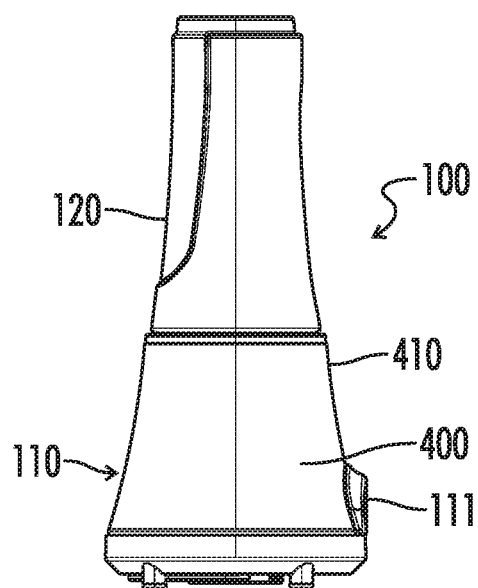


FIG. 3D

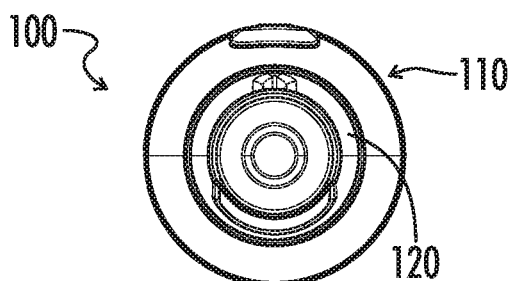


FIG. 3E

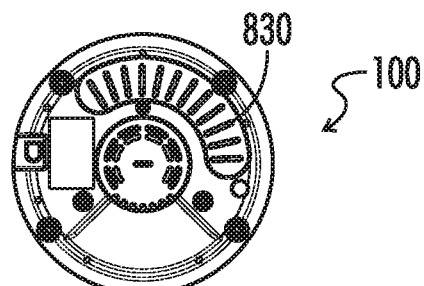


FIG. 3F

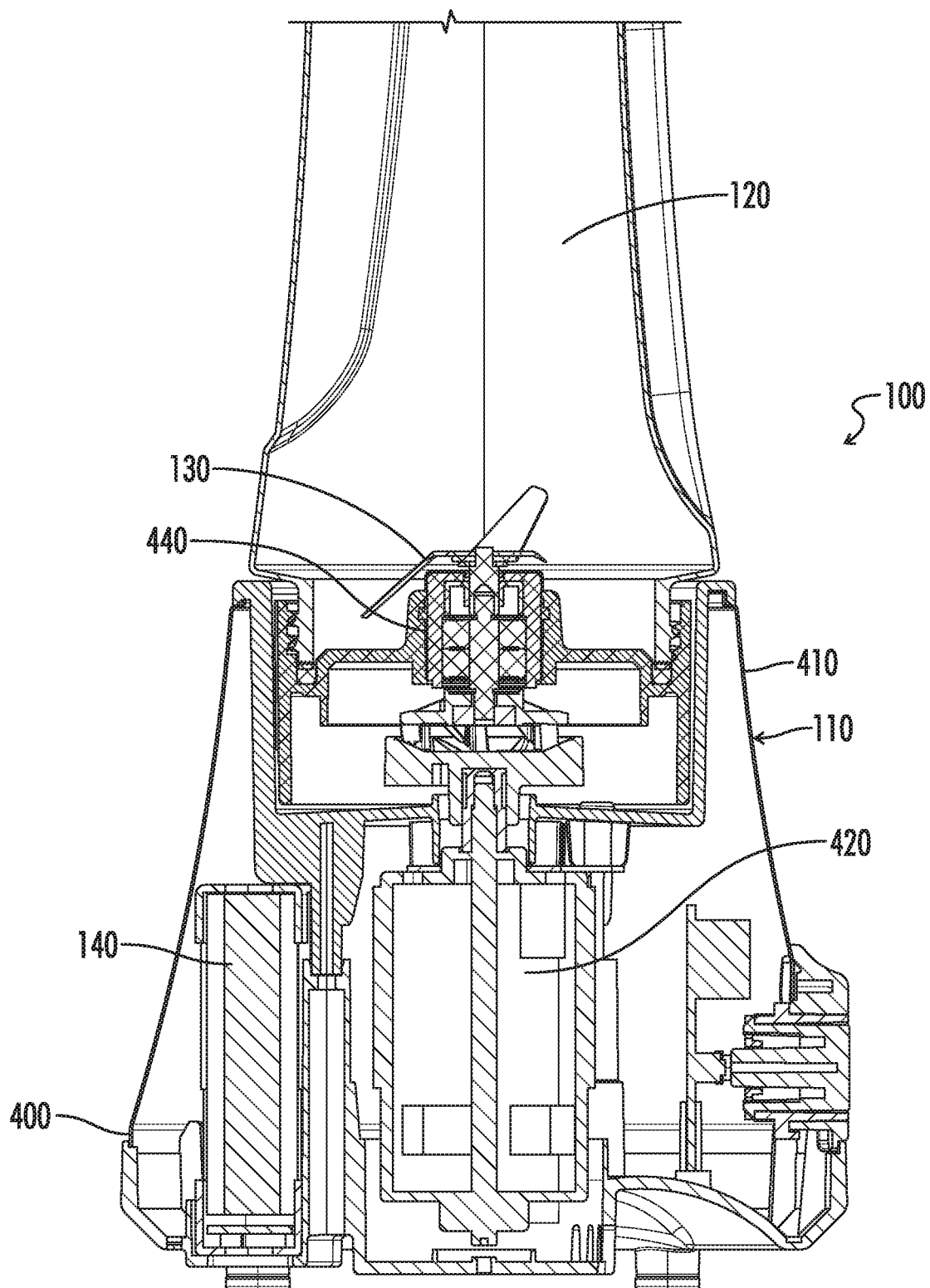


FIG. 4

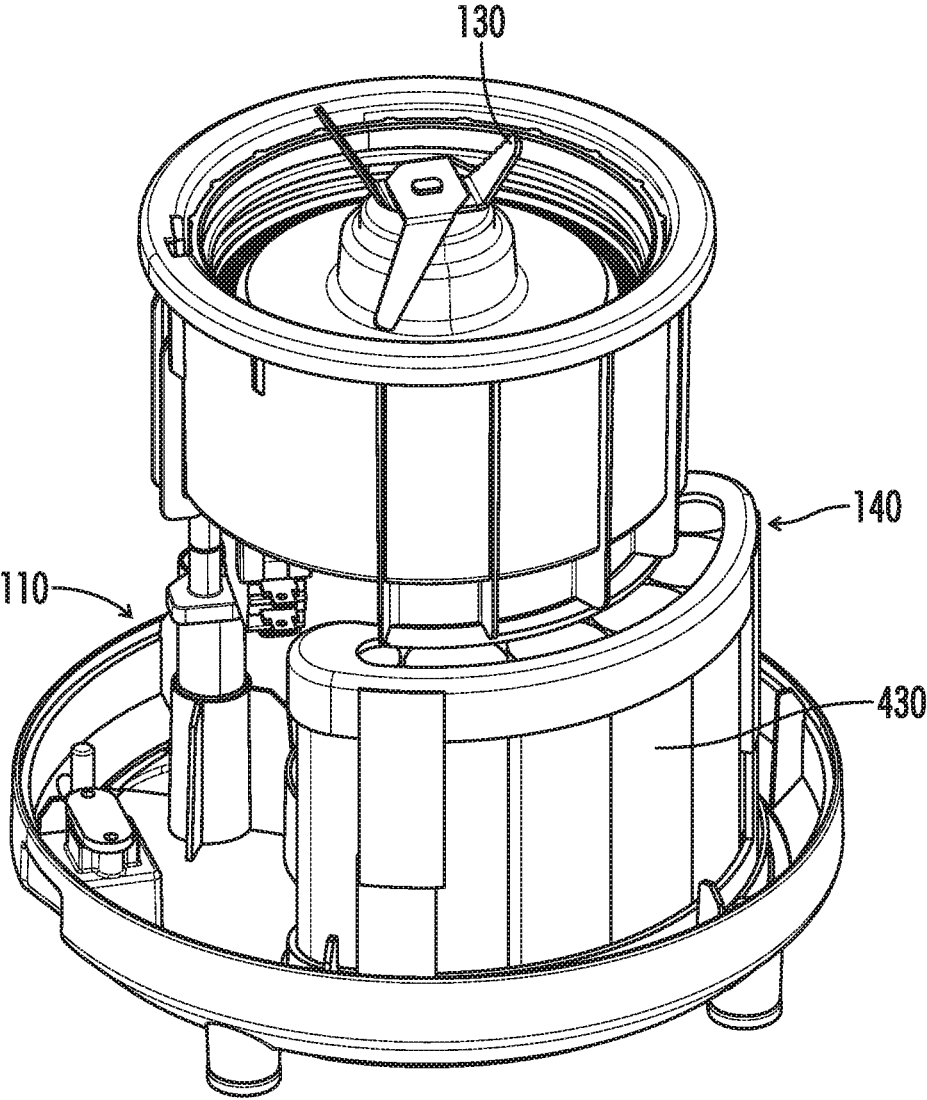


FIG. 5

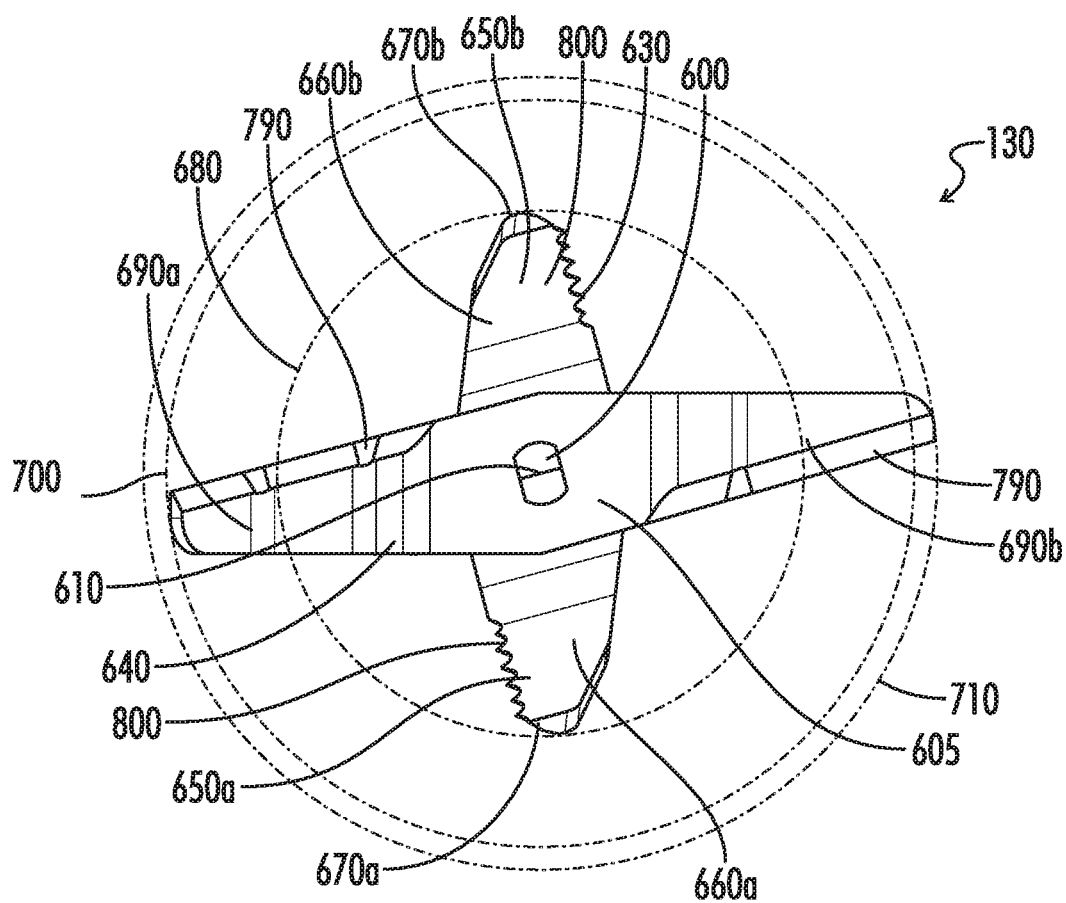


FIG. 6A

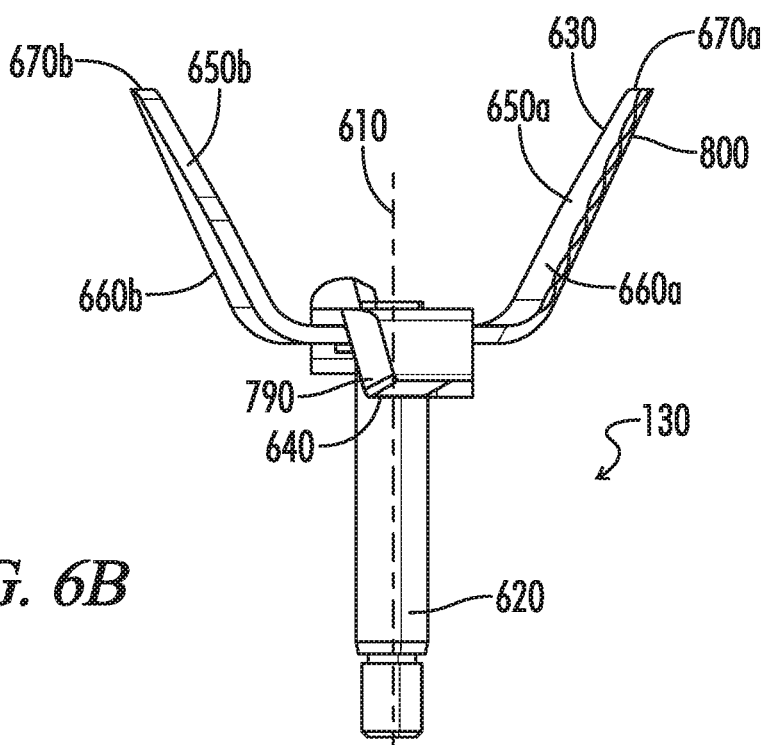


FIG. 6B

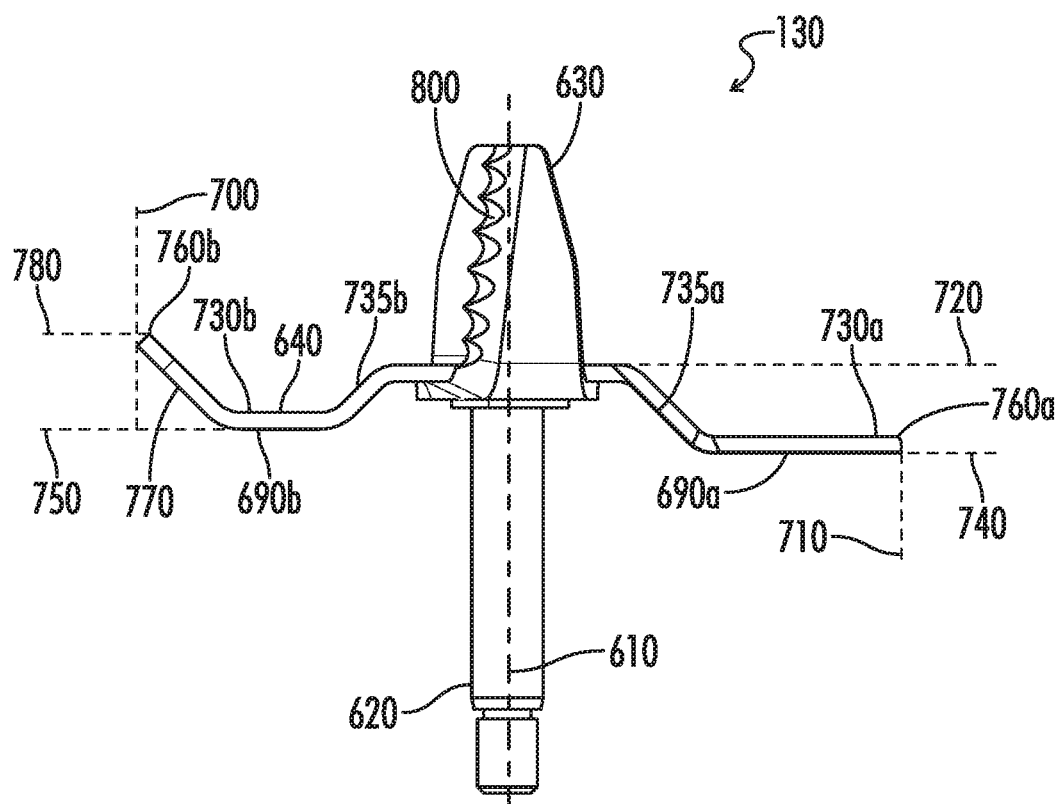


FIG. 6C

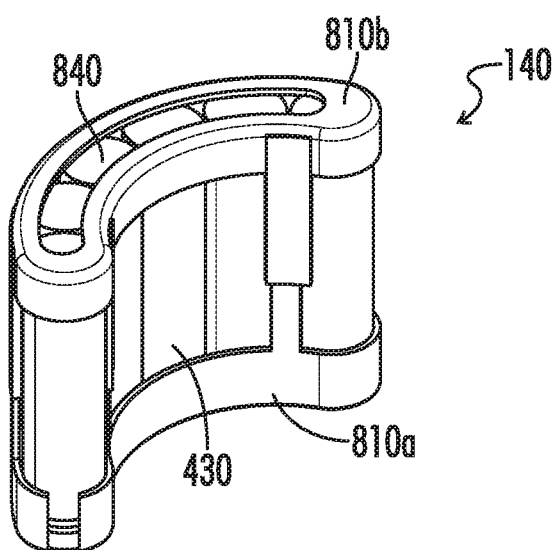


FIG. 7A

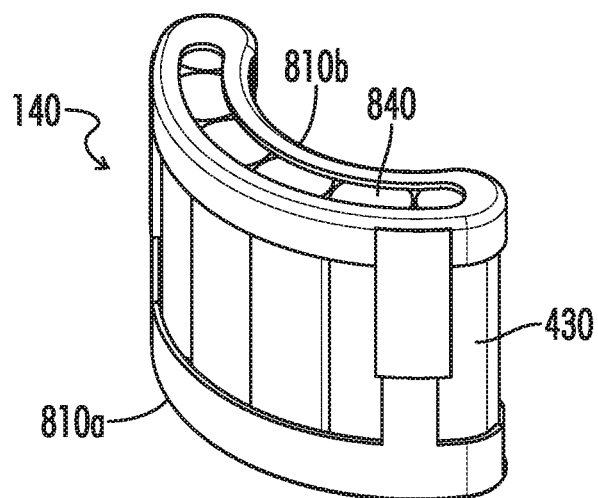


FIG. 7B

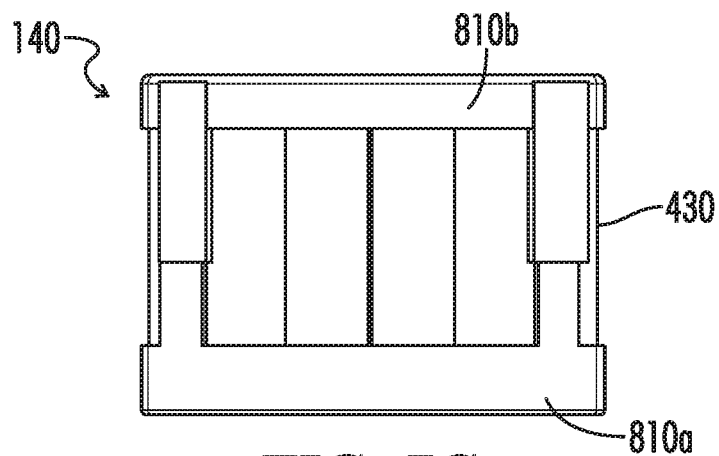


FIG. 7C

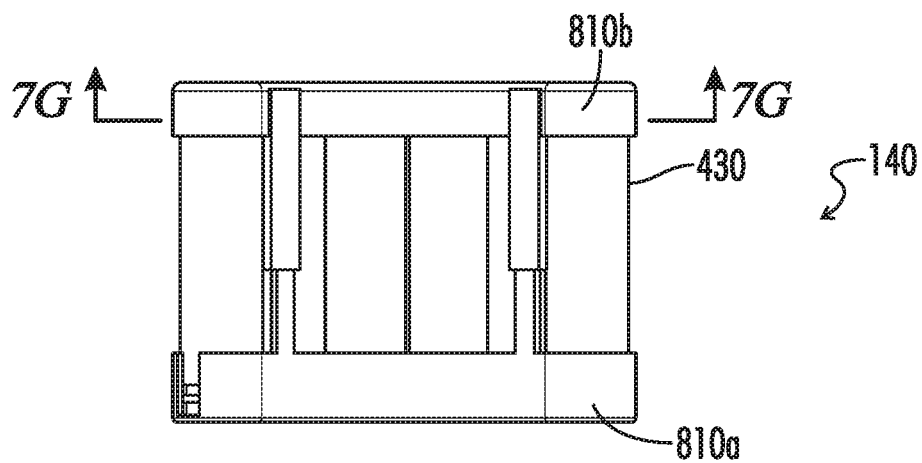


FIG. 7D

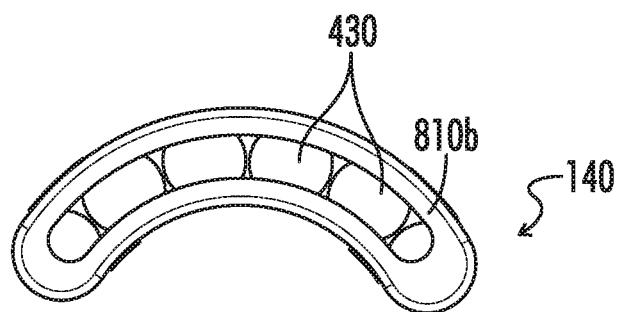


FIG. 7E

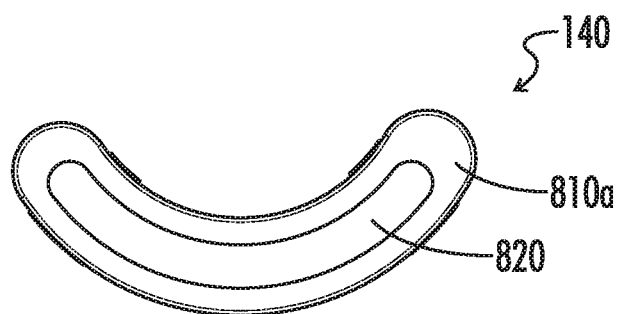


FIG. 7F

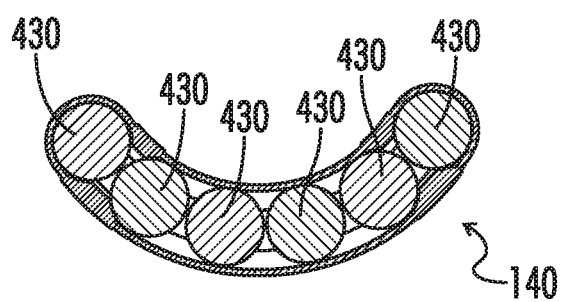


FIG. 7G

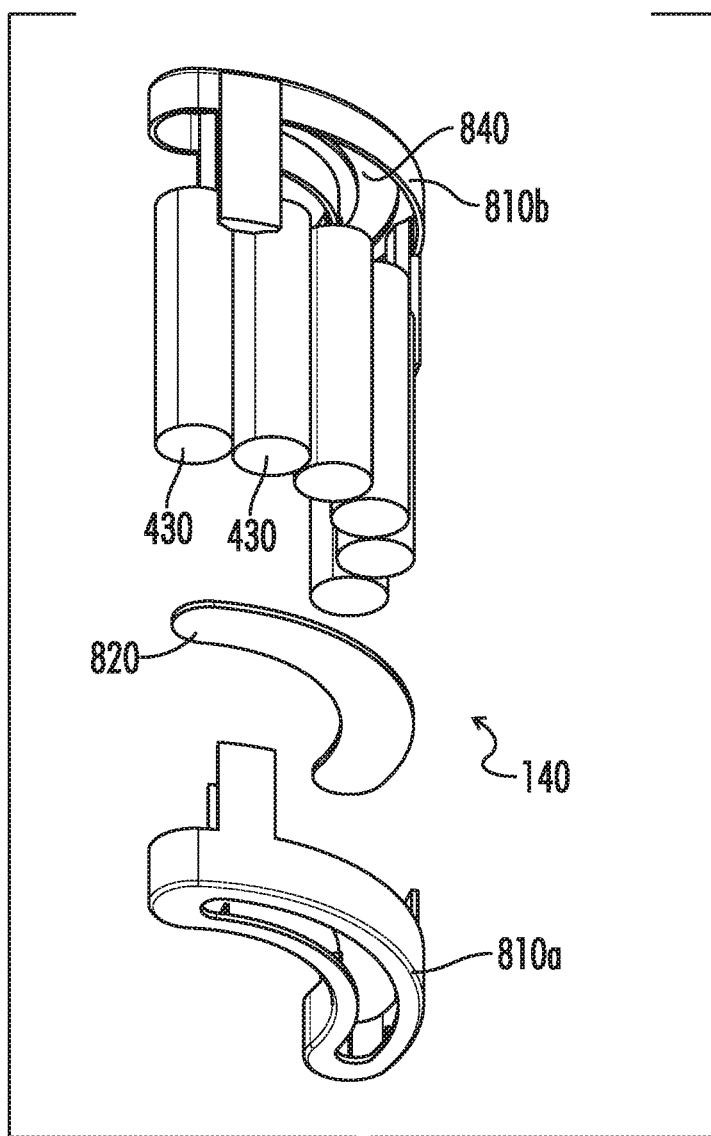


FIG. 7H

PORTABLE BLENDER

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims the benefit of U.S. Provisional Application No. 62/841,381, filed on May 1, 2019, the contents of which are incorporated by reference herein.

FIELD

[0002] The present disclosure generally relates to a portable blender.

BACKGROUND

[0003] Food and drink blenders are typically plugged into a power outlet in order to generate enough power to blend even the hardest consumable, such as frozen food and/or ice. The effectiveness of portable housewares and appliances that rely on battery power is usually related to the type and capacity of the battery. However, certain appliances are not particularly suited to being portable and transportable as the power requirement is usually too severe for effective operation.

[0004] Blenders typically require substantial amounts of power, and therefore portable blenders typically fail to function as well as plug in blenders. USB powered blenders exist, but typically lack sufficient battery capacity and use insufficient motors for power. Without a motor capable of delivering sufficient blending power, portable blenders cannot consistently break ice and other hard foods as well as plug in blenders can. Further, more powerful motors are not an option unless battery capacity can be increased.

[0005] There is a need for a portable blender that is portable and transportable and that can operate on enough power to blend even the hardest consumable, such as frozen food and/or ice.

SUMMARY

[0006] An embodiment of the present disclosure provides a portable blender, comprising: a base; a blade rotatably connected to the base; and a battery configured to supply power to rotate the blade with enough power to blend even the hardest consumable, such as frozen food and/or ice.

[0007] A portable blender according to this disclosure typically comprises a substantially cylindrical or conical base, a substantially cylindrical or conical blending chamber, and a blade assembly rotatably fixed within the blending chamber for blending items placed within the chamber.

[0008] A motor is provided within the base of the portable blender for rotating the blade assembly located within the blending chamber, and a battery assembly is provided within the base for powering the motor. The battery assembly comprises a plurality of substantially cylindrical battery cells arranged in a semicircular configuration and located adjacent an outer wall of the base.

[0009] In some embodiments, the base takes the form of a truncated cone and has a truncated conical surface, where the surface has a concave profile, thereby resulting in a bell shape.

[0010] In some embodiments, the battery assembly has an upper and lower bracket for positioning each of the battery cells in the semicircular configuration, and the lower bracket comprises a heat sink for cooling the battery cells. In such

an embodiment, the battery assembly is located above vents in a bottom surface of the base.

[0011] In some embodiments, the blade assembly comprises a fixation mechanism at a rotational axis of the blade assembly for fixing the blade assembly to a drive mechanism, a first pair of blades extending in opposite directions from the rotational axis, and a second pair of blades extending in opposite directions from the rotational axis.

[0012] The first pair of blades and the second pair of blades typically have different characteristics. The first pair of blades is typically symmetric about the rotational axis and the second pair of blades is typically asymmetric about the rotational axis.

[0013] The fixation mechanism is typically in a central planar blade segment, and the central planar blade segment is at a first height. The second pair of blades typically comprises a first blade having a first planar blade segment at a second height different than the first height and a second blade having a second planar blade segment at a third height different than the first height or the second height.

[0014] In some embodiments, the first planar blade segment ends in a blade tip and the second planar blade segment ends in an angled blade segment and the angled blade segment of the second blade ends in a blade tip.

[0015] The first blade and the second blade of the second pair may extend to different radial distances from the rotational axis. The blade tip of the first blade may therefore extend to a different radial distance than the blade tip of the second blade.

[0016] In some embodiments, the first pair of blades each have an angled blade segment angled upwards from the central blade segment, and each angled blade segment ends in a blade tip. The angled blade segment may then trace a radial circle when the blade assembly is rotated about the rotational axis. Each angled blade segment may then be canted relative to a tangent of the radial circle.

[0017] Further the radial circle traced by the blade tips of the first blade pair may be different than the radial circle traced by the first blade tip of the second pair or the radial circle traced by the second blade tip of the second pair. As such, the four blades of the blade assembly may trace three distinct radial circles.

[0018] In some embodiments, at least one blade of the blade assembly has a sharpened edge and at least one blade of the blade assembly has a serrated edge. In some such embodiments, the blades of the first pair have serrated blade edges and the blades of the second pair have sharpened edges.

[0019] In some embodiments, the blade assembly described is provided independent of the battery assembly described. Further, while the blade assembly is described in terms of a portable blender, the blade assembly may be used in a non-portable blender as well.

BRIEF DESCRIPTION OF THE DRAWINGS

[0020] FIGS. 1, 2A, 2B, and 2C illustrate perspective views of a portable blender according to one embodiment.

[0021] FIGS. 3A-3D illustrate side views of a portable blender according to one embodiment.

[0022] FIGS. 3E and 3F respectively illustrate a top view and a bottom view of a portable blender according to one embodiment.

[0023] FIG. 4 shows a sectioned view of the base of the portable blender of FIG. 1 with the container attached.

[0024] FIG. 5 shows a view of the base of the portable blender of FIG. 1 with a housing removed.

[0025] FIGS. 6A-C show top and side views of a blade assembly for use in the portable blender of FIG. 1.

[0026] FIGS. 7A-F show perspective, front, back, top, and bottom views of a battery assembly for use in the portable blender of FIG. 1.

[0027] FIG. 7G shows a sectioned top view of the battery assembly of FIG. 7A.

[0028] FIG. 7H shows an exploded view of the battery assembly of FIG. 7A.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0029] The description of illustrative embodiments according to principles of the present disclosure is intended to be read in connection with the accompanying drawings, which are to be considered part of the entire written description. In the description of embodiments of the disclosure disclosed herein, any reference to direction or orientation is merely intended for convenience of description and is not intended in any way to limit the scope of the present disclosure. Relative terms such as “lower,” “upper,” “horizontal,” “vertical,” “above,” “below,” “up,” “down,” “top” and “bottom” as well as derivative thereof (e.g., “horizontally,” “downwardly,” “upwardly,” etc.) should be construed to refer to the orientation as then described or as shown in the drawing under discussion. These relative terms are for convenience of description only and do not require that the apparatus be constructed or operated in a particular orientation unless explicitly indicated as such. Terms such as “attached,” “affixed,” “connected,” “coupled,” “interconnected,” and similar refer to a relationship wherein structures are secured or attached to one another either directly or indirectly through intervening structures, as well as both movable or rigid attachments or relationships, unless expressly described otherwise. Moreover, the features and benefits of the disclosure are illustrated by reference to the exemplified embodiments. Accordingly, the disclosure expressly should not be limited to such exemplary embodiments illustrating some possible non-limiting combination of features that may exist alone or in other combinations of features; the scope of the disclosure being defined by the claims appended hereto.

[0030] This disclosure describes the best mode or modes of practicing the disclosure as presently contemplated. This description is not intended to be understood in a limiting sense, but provides an example of the disclosure presented solely for illustrative purposes by reference to the accompanying drawings to advise one of ordinary skill in the art of the advantages and construction of the disclosure. In the various views of the drawings, like reference characters designate like or similar parts.

[0031] It is important to note that the embodiments disclosed are only examples of the many advantageous uses of the innovative teachings herein. In general, statements made in the specification of the present application do not necessarily limit any of the various claimed disclosures. Moreover, some statements may apply to some inventive features but not to others. In general, unless otherwise indicated, singular elements may be in plural and vice versa with no loss of generality.

[0032] FIGS. 1, 2A, 2B, and 2C illustrate perspective views of a portable blender 100 according to one embodi-

ment. FIGS. 3A-3D illustrate side views of the portable blender 100. FIGS. 3E and 3F respectively illustrate a top view and a bottom view of the portable blender 100.

[0033] The portable blender 100 shown in FIG. 1 may be used to crush, chop, mix, shred, or blend material. In particular, the portable blender 100 may have a unique battery assembly 140 and blade assembly 130 that allows the blender 100 to crush even the hardest consumables, such as frozen food and/or ice. The portable blender 100 may include a base 110, a container 120, and the blade assembly 130. The base 110 may include an on/off switch 111 for rotation of the blade assembly 130. The switch 111 may be disposed on a side of the base 110. The blade assembly 130 may be rotatably connected to the base 110. The container 120 may be configured to attach to the base 110, and may accommodate the blade assembly 130.

[0034] The container 120 may be configured to hold material to be chopped, mixed, shredded or blended by the blade assembly 130. The material may include, but not limited to, fruits, vegetables, nuts, grains, meats, eggs, cheese, spices, other food or food products, drinkable liquids, frozen food, ice, or other edible ingredients.

[0035] FIG. 4 shows a sectioned view of the base 110 of the portable blender 100 of FIG. 1 with the container 120 attached. FIG. 5 shows a perspective view of the base 110 of the portable blender 100 of FIG. 1 with the housing removed.

[0036] The portable blender 100 typically comprises a substantially cylindrical or conical base 110, or a base shaped in some combination thereof. For example, the base 110 may be bell shaped, as in the embodiment shown. As shown, the base 110 has a housing 400 which may take the form of a truncated cone. As such a surface 410 of the truncated conical housing 400 may have a concave profile.

[0037] The portable blender 100 typically further comprises a blending chamber, or container 120, which may be similarly cylindrical or conical. In some embodiments, such as that shown, the housing 400 of the base 110 and the container may form a single cylindrical or conical shape, including an extended bell shape, when combined.

[0038] The portable blender 100 further comprises a blade assembly 130 rotatably fixed within the blending chamber 120 and a motor 420 for rotating the blade assembly 130 is located within the base 110. Further, the battery assembly 140 for powering the motor 420 is fixed within the base 110. As shown, and as discussed in more detail below, the battery assembly 140 for powering the motor 420 comprises a plurality of substantially cylindrical cells 430 arranged in a semicircular configuration. The battery assembly 140 is then located adjacent an outside wall 410 of the base 110 of the portable blender 100.

[0039] When in use, the blending chamber 120 is removably fixed to the base 110 of the portable blender 100. When the blending chamber 120 is on the base 110, the motor 420 in the base rotates the blade assembly 130 by way of a linkage 440.

[0040] FIGS. 6A-C show top and side views of a blade assembly 130 for use in the portable blender 100 of FIG. 1. The blade assembly 130 has a fixation mechanism 600 located at a rotational axis 610 of the blade assembly for fixing the blade assembly to a drive mechanism. The fixation mechanism 600 may be a hole at a center point of the blade assembly, with the hole being shaped to accept a drive shaft 620. Alternatively, the drive shaft 620 may be permanently

fixed to the blade assembly **130**, and may form part of the fixation mechanism **600**. The fixation mechanism **600** may then be rotated by way of the linkage **440**.

[0041] The fixation mechanism **600** is typically located centrally at a central planar blade segment **605**. The blade assembly **130** has a first pair of blades **630** extending in opposite directions from the rotational axis **610** and a second pair of blades **640** extending in opposite directions from the rotational axis. The second pair of blades **640** are substantially perpendicular to the first pair of blades **630**. Generally, as shown, the first pair of blades **630** are symmetric while the second pair of blades **640** are asymmetric about the rotational axis **610**.

[0042] The first pair of blades **630** typically has a first blade **650a** and a second blade **650b** which are symmetric about the rotational axis **610**. As shown, the blades **650a, b** of the first pair of blades **630** each have an angled blade segment **660a, b** angled upwards from the central blade segment **605**. The angled blade segment ends in a blade tip **670a, b**, and upon rotation of the blade assembly **130** about the rotational axis **610**, the blade tips trace a circle **680** at a first radial distance from the rotational axis.

[0043] As shown, the angled blade segment **660a, b** of each of the blades **650a, b** of the first pair **630** are canted relative to a tangent of the circle **680** at the first radial distance. This results in a force pushing any food products in the container **120** outwards during rotation of the first pair of blades **630**. Typically, the angled blade segments **660a, b** are canted 15-16 degrees relative to the tangent of the circle **680**. However, more aggressive or less aggressive cant angles are possible.

[0044] The second pair of blades **640** has a first blade **690a** and a second blade **690b** extending from the central planar blade segment **605**. As shown, the blades **690a, b** of the second pair **640** are asymmetric, and as such the first blade **690a** traces a circle **700** at a second radial distance from the rotational axis **610** and the second blade **690b** traces a circle **710** at a third radial distance from the rotational axis. The first, second, and third radial distances are different, and as such the circles **700, 710** traced by the two blades **690a, b** of the second pair **640** are different than the circle **680** traced by the blades **650a, b** of the first pair **630**.

[0045] As shown in FIG. 6C, among others, the fixation mechanism **600** is located in the central planar blade segment **605** and is located at a first height **720**. The blades **690a, b** of the second pair **640** each have a respective planar blade segment **730a, b**, and the planar blade segment **730a** of the first blade **690a** is at a height **740** different than a height **750** of the planar blade segment **730b** of the second blade **690b**. Further, the heights **740, 750** of the planar blade segments **730a, b** of the second pair **640** are different than the height **720** of the central planar blade segment **605**. In this way, the three planar blade segments **605, 730a, b** are at three distinct heights **720, 740, 750**. As shown, the heights **740, 750** of the planar blade segments **730a, b** may be below the height **720** of the central planar blade segment **605**.

[0046] Further, the first blade **690a** of the second pair **640** has a transitional angled blade segment **735a** between the central planar blade segment **605** and the first planar blade segment **730a**. Similarly, the second blade **690b** of the second pair **640** has a transitional angled blade segment **735b** between the central planar blade segment **605** and the second planar blade segment **730b**. As shown, the first angled blade segment **735a** and the second angled blade

segment **735b** may be at the same angle relative to the planar blade segments **605, 730a, b**. As shown, the second planar blade segment **730b** is at a height **750** higher than the first planar blade segment **730a**, such that the height **750** of the second planar blade segment **730b** is closer to the height **720** of the central planar blade segment **605** than the height **740** of the first planar blade segment **730a**. In such an embodiment, the second planar blade segment **730b** may begin further radially from the central planar blade segment **605** than the first planar blade segment **730a**.

[0047] Further, as shown, the planar blade segment **730a** of the first blade **690a** ends in a blade tip **760a**. In contrast, the planar blade segment **730b** of the second blade **690b** ends in an angled blade segment **770**, and the angled blade segment ends in a blade tip **760b**. In this way, the blade tip **760b** of the second blade **690b** is at a height **780** different than the first height **720**, second height **730**, or the third height **740**. As discussed above, the blade tip **760a** of the first blade **690a** extends to a distance radially than the blade tip **760b** of the second blade, and both blade tips **690a, b** of the second pair **640** extend to different distances radially than either blade tip **670a, b** of the first pair.

[0048] In some embodiments, at least one blade has a sharpened blade edge **790** and at least one blade has a serrated blade edge **800**. In the embodiment shown, each blade **650a, b** of the first pair **630** has a serrated blade edge **800** and each blade **690a, b** of the second blade pair **640** has a sharpened blade edge **790**.

[0049] FIGS. 7A-F show perspective, front, back, top, and bottom views of a battery assembly **140** for use in the portable blender **100** of FIG. 1. FIG. 7G shows a sectioned top view of the battery assembly **140** of FIG. 7A, and FIG. 7H shows an exploded view of the battery assembly **140**.

[0050] As shown, the battery assembly **140** has a plurality of substantially cylindrical battery cells **430** arranged in a semicircular configuration. The battery assembly **140** typically has an upper bracket **810a** and a lower bracket **810b** which combine to position each of the battery cells **420** in the semicircular configuration.

[0051] The lower bracket **810b** may include circuitry **820** for managing the charging and discharging of each of the battery cells **420**, and may further include a heat sink mechanism for directing heat generated by the cells **420** downwards. As such, the lower bracket **810b** may be positioned directly above vents **830** in a bottom surface of the base **110** of the portable blender **100**, and heat may be directed out the vents.

[0052] The upper bracket **810b** may comprise an opening **840** exposing an end of each of the battery cells **420**.

[0053] The semicircular configuration applied to the individual battery cells **420** by the upper and lower brackets **810a, b** allows the battery assembly **130** to be located directly adjacent the cylindrical or conical outer walls **410** of the housing **400** of the base **110**. Otherwise, significant dead space would exist between a traditionally shaped battery and the outer walls **410**, and a smaller capacity battery would be required to fit a designated battery compartment space. The curved shape of the battery assembly **130** thereby allows for a higher capacity battery which can thereby power a more powerful motor **420** in the portable blender **100**.

[0054] Because of the additional battery capacity made possible by the configuration shown, the motor **420** may have a higher motor speed, such as 15,500 RPM, allowing for quick and efficient blending of frozen foods. The motor

may have a double ball bearing structure design, and may have a high efficiency (greater than 70%), thereby improving motor frequency and overall motor life and reducing power consumption and extending life of the battery pack.

[0055] While the present disclosure has been described at some length and with some particularity with respect to the several described embodiments, it is not intended that it should be limited to any such particulars or embodiments or any particular embodiment, but it is to be construed with references to the appended claims so as to provide the broadest possible interpretation of such claims in view of the prior art and, therefore, to effectively encompass the intended scope of the disclosure.

[0056] All examples and conditional language recited herein are intended for pedagogical purposes to aid the reader in understanding the principles of the disclosure and the concepts contributed by the inventor to furthering the art, and are to be construed as being without limitation to such specifically recited examples and conditions. Moreover, all statements herein reciting principles, aspects, and embodiments of the disclosure, as well as specific examples thereof, are intended to encompass both structural and functional equivalents thereof. Additionally, it is intended that such equivalents include both currently known equivalents as well as equivalents developed in the future, i.e., any elements developed that perform the same function, regardless of structure.

What is claimed is:

1. A portable blender, comprising:
 - a substantially cylindrical or conical base;
 - a substantially cylindrical or conical blending chamber;
 - a blade assembly rotatably fixed within the blending chamber;
 - a motor for rotating the blade assembly located within the base; and
 - a battery assembly for powering the motor located within the base,
 wherein the battery assembly comprises a plurality of substantially cylindrical battery cells arranged in a semicircular configuration and located adjacent an outer wall of the base.
2. The portable blender of claim 1, wherein the base is has a substantially conical and truncated surface, and wherein surface has a concave profile resulting in a partial bell shape.
3. The portable blender of claim 1, wherein the battery assembly further comprises an upper and lower bracket for positioning each of the battery cells in the semicircular configuration.
4. The portable blender of claim 3, wherein the lower bracket comprises a heat sink for cooling the battery cells, and wherein the battery assembly is located above vents in a bottom surface of the base.
5. The portable blender of claim 1, wherein the blade assembly comprises:
 - a fixation mechanism at a rotational axis of the blade assembly for fixing the blade assembly to a drive mechanism;
 - a first pair of blades extending in opposite directions from the rotational axis;
 - a second pair of blades extending in opposite directions from the rotational axis, the second pair of blades being substantially perpendicular to the first pair of blades,

wherein the first pair of blades is symmetric and the second pair of blades is not symmetric about the rotational axis.

6. The portable blender of claim 5, wherein the second pair of blades comprises a first blade and a second blade extending to different radial distances from the rotational axis.

7. The portable blender of claim 5, wherein the fixation mechanism is in a central planar blade segment at a first height, and the second pair of blades comprises a first blade having a first planar blade segment at a second height different than the first height and a second blade having a second planar blade segment at a third height different than the first height or the second height.

8. The portable blender of claim 7, wherein the first planar blade segment ends in a blade tip and the second planar blade segment ends in an angled blade segment, and where the angled blade segment ends in a blade tip.

9. The portable blender of claim 8, wherein the blade tip of the first blade of the second pair extends to a first radial distance and the blade tip of the second blade of the second pair extends to a second radial distance different than the first radial distance.

10. The portable blender of claim 5, wherein the fixation mechanism is in a central blade segment, and wherein the first pair of blades each have an angled blade segment angled upwards from the central blade segment, each angled blade segment ending in a blade tip, and wherein the angled blade segment traces a radial circle when the blade assembly is rotated about the rotational axis, and wherein the angled blade segment is canted relative to a tangent of the radial circle.

11. The portable blender of claim 5, wherein at least one blade has a sharpened blade edge and at least one blade has a serrated blade edge.

12. The portable blender of claim 11, wherein the first pair of blades comprise serrated blade edges and wherein the second pair of blades comprise sharpened blade edges.

13. A blade assembly for a blender comprises:

- a fixation mechanism at a rotational axis of the blade assembly for fixing the blade assembly to a drive mechanism;
- a first pair of blades extending in opposite directions from the rotational axis;
- a second pair of blades extending in opposite directions from the rotational axis, the second pair of blades being substantially perpendicular to the first pair of blades, wherein the first pair of blades is symmetric and the second pair of blades is not symmetric about the rotational axis.

14. The blade assembly of claim 13, wherein the second pair of blades comprises a first blade and a second blade extending to different radial distances from the rotational axis.

15. The blade assembly of claim 13, wherein the fixation mechanism is in a central planar blade segment at a first height, and the second pair of blades comprises a first blade having a first planar blade segment at a second height different than the first height and a second blade having a second planar blade segment at a third height different than the first height or the second height.

16. The blade assembly of claim 15, wherein the first planar blade segment ends in a blade tip and the second

planar blade segment ends in an angled blade segment, and where the angled blade segment ends in a blade tip.

17. The blade assembly of claim **16**, wherein the blade tip of the first blade of the second pair extends to a first radial distance and the blade tip of the second blade of the second pair extends to a second radial distance different than the first radial distance.

18. The blade assembly of claim **13**, wherein the fixation mechanism is in a central blade segment, and wherein the first pair of blades each have an angled blade segment angled upwards from the central blade segment, each angled blade segment ending in a blade tip, and wherein the angled blade segment traces a radial circle when the blade assembly is rotated about the rotational axis, and wherein the angled blade segment is canted relative to a tangent of the radial circle.

19. The blade assembly of claim **13**, wherein at least one blade has a sharpened blade edge and at least one blade has a serrated blade edge.

20. The blade assembly of claim **19**, wherein the first pair of blades comprises serrated blade edges and wherein the second pair of blades comprises sharpened blade edges.

* * * * *